

WASP-10b

R-1580

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_140927 · created 2026-08-01T23:50:08+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456927.9187 +/- 0.0024 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.196 +/- 0.057
Transit depth (Rp/Rs)^2	3.85 +/- 2.25 [%]
Orbital Inclination (inc)	85.5 +/- 1.82
Airmass coefficient 1 (a1)	0.932 +/- 0.042
Airmass coefficient 2 (a2)	0.038 +/- 0.021
Scatter in the residuals of the lightcurve fit is	4.48 %
Best Comparison Star	#3 - [625, 232]
Optimal Aperture	3.6
Optimal Annulus	7.49
Transit Duration (day)	0.055 +/- 0.034

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.196 ± 0.057 vs 0.1592 (deviation 0.65σ)
Duration fit vs published	0.055 d vs 0.0946 d (deviation 1.16σ)
Mid-time O–C	5.2 min
Depth SNR	3.4
Residual scatter	4.48 %

Light curve

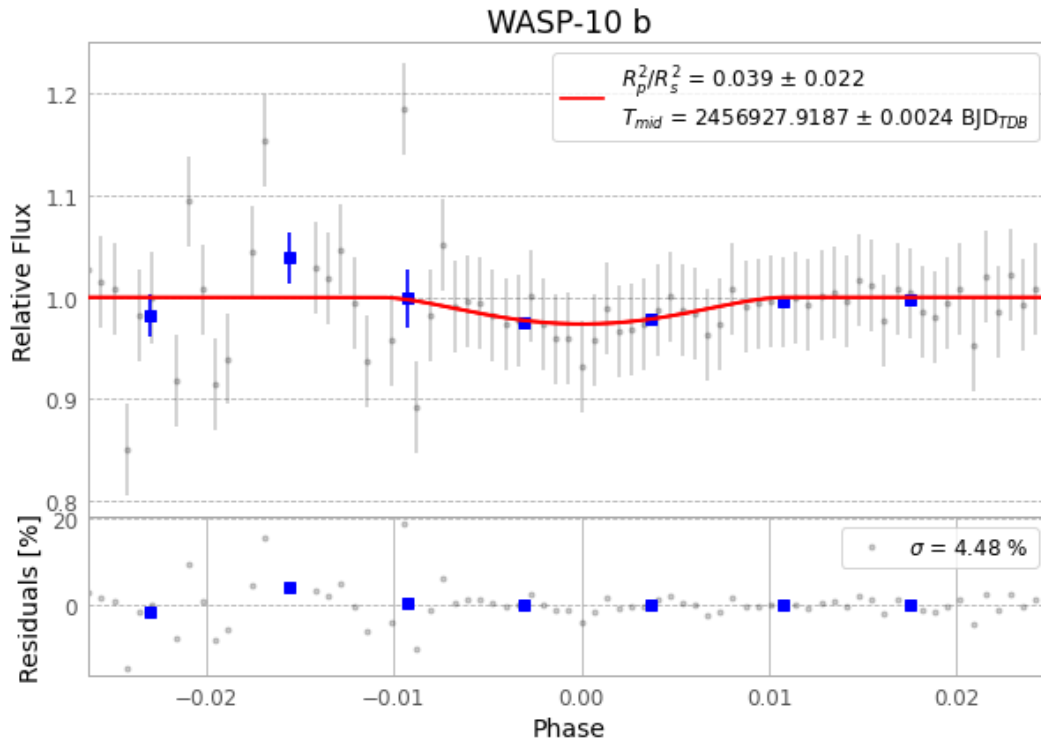


Figure 1 — FinalLightCurve_WASP-10 b_2014-09-27.png (SHA-256 e687fe87e7de8a5d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [280, 273], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d1f579dadcebebd8...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-10140927075712.FITS → WASP-10140927114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-140927.log (ede8112530bb...), FinalLightCurve_WASP-10 b_2014-09-27.png (e687fe87e7de...), FinalLightCurve_WASP-10 b_2014-09-27.csv (a5762fd897ba...)

Compute resources

Wall time 135.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-01 23:50Z.